

3-Mark Questions & Answers — MPCA Module 5: 80386DX & Pentium Processors

Q1. Explain the 4 privilege rings of 80386 Protected Mode.

- **Ring 0 (Kernel Core):** Highest privilege; executes privileged OS instructions.
 - **Ring 1 (Device Drivers):** System services and drivers.
 - **Ring 2 (OS Extensions):** Middleware and database engines.
 - **Ring 3 (User Applications):** Lowest privilege; untrusted user applications.
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Q2. Explain the 5 pipeline stages of Pentium processor.

1. **Prefetch (PF):** Fetches instructions from 8 KB Code Cache.
 2. **Decode-1 (D1):** Decodes opcode and determines U/V pairability.
 3. **Decode-2 (D2):** Generates memory addresses for operands.
 4. **Execute (EX):** ALU integer execution and Data Cache access.
 5. **Writeback (WB):** Updates destination registers and flags.
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Q3. Explain two-level Paging in 80386 microprocessor.

Linear address (32-bit) is divided into:

- **Bits 31-22 (10 bits)** : Page Directory Index.
 - **Bits 21-12 (10 bits)** : Page Table Index.
 - **Bits 11-0 (12 bits)** : Offset within 4 KB Page Frame.
 - **CR3** register holds the physical base address of the Page Directory.
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